RM10: Recovery/acceleration, information, and useful ramp gradients

Methodology review — Option A only

5 October 2026

Conclusion. Physical acceleration/recovery has support. A positive task with genuinely hidden, action-relevant information is not admitted in this bounded review. Conditional aggregate sufficiency and missing rich-family evidence are different findings. Option C was not opened.

1. Task and complete population accounting

This is an audit template, not an adopted traffic update. Let $s = 0, \dots, S$ index a common time grid with step Δt (seconds). Let $\theta \in \mathbb{R}^d$ parameterize a bounded common open-loop ramp rate $u_s(\theta)$ (vehicle-equivalents per second). The state x_s contains road inventories, ramp/external queues and whatever recovery variables a justified model requires. Public parameters are p and public arrivals/boundary inputs are b_s. A lawful discretization would give

$$x_{s+1} = F_s(x_s, u_s(\theta), b_s; p).$$

Permission u_s, release at a meter, and actual entry at a merge are distinct: queues, travel and receiving can prevent permission being realized. This template specifies no new release or merge rule.

Let $P_s = P(x_s)$ be *all* arrived but not completed traffic mass, including external waiting and both road and ramp traffic. Let a_s be external demand arrival rate and e_s be final scenario exit rate. Internal transfers cancel:

$$P_{s+1} = P_s + \Delta t(a_s - e_s), \qquad J(\theta) = \Delta t \sum_{s=0}^{S-1} P_s(\theta). \tag{1}$$

J has vehicle-equivalent seconds as units; J/M is mean system time for a common fixed total demand mass M. The grid convention is left-endpoint time accounting. A common S must clear *all* comparison policies and small perturbations; truncation with traffic remaining is not full TST.

Iterating the balance gives

$$P_s = P_0 + \Delta t \sum_{j=0}^{s-1} (a_j - e_j).$$

For fixed P_0, a_j and fixed S, therefore,

$$\nabla_\theta J = -\Delta t^2 \sum_{j=0}^{S-2} (S - 1 - j) \nabla_\theta e_j. \tag{2}$$

Earlier exits reduce more subsequent waiting intervals. Merely moving a queue from a mainline lane to the ramp leaves this objective unchanged when final exit history is unchanged. Mean steady capacity is not the exit-time history.

2. What hidden information must mean

Let O be a declared observation map with spatial and temporal support. A candidate history through decision epoch t consists of

$$H_t = (O(x_0), \dots, O(x_t), u_{0:t-1}, b_{0:t-1}, p).$$

It includes admitted inventories, completed face counts/flows, speed and ramp observations. Public future demand may also be common; future observations are not available before selecting an action.

If N_i is cell mass and actual speed observations are a section mass mean, the idealized definition is

$$N_i = \sum_k m_{ik}, \qquad U_i = \frac{\sum_k m_{ik} v_{ik}}{N_i}, \quad N_i > 0.$$

m_{ik} has vehicle-equivalent units and v_{ik}, U_i have speed units; k indexes traffic groups within cell i. Empty-cell speed is unavailable. This illustrates support, not approval of a new sensor contract. Face/time speed means and equilibrium FD proxies differ from U_i. Do not use the closed RM9 FD-speed proxy as an actual non-equilibrium speed law.

Define the consistency set using source-lawful initial states $\mathcal{D}_0$:

$$\mathcal{X}(H_t) = \left\{ x_t : \begin{array}{l} x_0 \in \mathcal{D}_0, \quad x_{s+1} = F_s(x_s, u_s, b_s; p) \ (0 \le s < t), \\ O(x_s) = H_t[s] \ (0 \le s \le t) \end{array} \right\}.$$

Different snapshots with equal total mass do not establish different future-relevant states in this set. Match public traffic characteristics and forcing; changing a random future realization is not a hidden initial-state certificate.

Conditional sufficiency. In a well-defined deterministic model, a known complete x_0 and public inputs determine every x_s by induction. Likewise, if history identifies a complete state at an earlier epoch, later congestion does not erase that knowledge. This is conditional on completeness and uniqueness; an existence result for a PDE alone is not such a uniqueness proof.

For a homogeneous aggregate model with complete state (N_i, U_i) and observed queues, exact observation observes that complete state. This does not establish physical adequacy or identifiability of unobserved lane speeds, within-cell waves, leaders or distributions.

A sufficient action certificate. For two source-lawful states $x_t^a, x_t^b \in \mathcal{X}(H_t)$ and two common feasible rate schedules θ_1, θ_2, opposite TST rankings demonstrate action relevance:

$$J_a(\theta_1) < J_a(\theta_2), \qquad J_b(\theta_1) > J_b(\theta_2).$$

This distinguishes these two actions; it does not prove that all possible optimal action sets are disjoint. Opposite useful local descent directions could provide another certificate. Neither is established here.

3. Why recovery alone does not prove necessary retained state

An inspected density-based aggregate competitor is Srivastava's queue-discharge demand law (2016 dissertation, Chapter 3.3.1, equation 3.3):

$$D_{\mathrm{rec}}(k) = \min\{v_f k,\ q_j + c_*(k_j - k)\}, \qquad c_* = \frac{q_c - q_j}{k_j - k_c}.$$

k, k_j, k_c are density, jam density and critical density; v_f is free-flow speed; q_c is capacity; q_j is jam demand; c_* has speed units. This source changes demand while retaining ordinary supply and conservation. It is an aggregate recovery precedent for queue discharge, not a validated ramp-merge law imported by this review.

Another competitor retains a hysteresis mode σ_s. With public thresholds $k_- < k_+$ and a known initial mode, a representative recurrence is

$$\sigma_{s+1} = \begin{cases} 1, & k_{s+1} \geq k_+, \\ 0, & k_{s+1} \leq k_-, \\ \sigma_s, & k_- < k_{s+1} < k_+. \end{cases}$$

This is the known-history logic of Cao/Nilsson/Coogan (2021), Section II.B, with indexing restated. Complete density history reconstructs this mode. An unknown initial mode is a different, unresolved domain.

For richer recovery, N_i and U_i constrain only the zeroth and first moments of a speed distribution ν_i:

$$N_i = \int d\nu_i(v), \qquad N_i U_i = \int v\, d\nu_i(v).$$

These two constraints generally do not specify ν_i. This algebra does *not* prove that distinct distributions are source-lawful, persist under the same full history, or change useful ramp-rate actions.

The primary studies support hybrid/local recovery and mean-capacity competitors. In particular, S12A005's similar mean capacities do not imply exact transient closure. Its differing transient flow traces are not identical observed histories. Laurent-Brouty/Costeseque/Goatin's macroscopic model retains local moving constraints: the label macro cannot be used to dismiss retained state.

Completed screening conclusion. Closed density/rate laws, known history modes, and completely observed homogeneous speed-state models have conditional explanations for sufficiency. Rich distribution/leader models retain physical potential, but lack a joined lawful initialization, equal-history and useful-action certificate for this task. Negative admission concerns this review envelope and evidence, not every bounded-acceleration model.

4. General gradient chain and its limits

The intended path is

$$\begin{aligned}\theta &\longrightarrow \text{permission/release} \longrightarrow \text{actual entry/recovery}\\ &\longrightarrow \text{receiving/conservation} \longrightarrow \text{final exits/TST.}\end{aligned}$$

Consider a chosen source-lawful discretization on a locally differentiable branch. Fold the public rate parameterization into $\widetilde{F}_s(x,\theta) = F_s(x, u_s(\theta), b_s; p)$. Define

$$A_s = \partial_x \widetilde{F}_s, \quad B_s = \partial_\theta \widetilde{F}_s, \quad S_s = \partial_\theta x_s.$$

For a state of dimension n, $A_s \in \mathbb{R}^{n\times n}$ and $B_s, S_s \in \mathbb{R}^{n\times d}$. If x_0 is fixed independently of θ,

$$\begin{aligned}S_0 &= 0,\\ S_{s+1} &= A_s S_s + B_s,\\ \nabla_\theta J &= \Delta t \sum_{s=0}^{S-1} S_s^\top \nabla_x P(x_s).\end{aligned}$$

These are chain-rule identities, not a numerical correctness result. They agree with the exit-balance expression when both use the same update and complete population. Normalized gradients divide by the common M.

The corresponding adjoint, with no terminal cost and fixed S, is

$$\lambda_S = 0, \qquad \lambda_s = \Delta t \nabla_x P(x_s) + A_s^\top \lambda_{s+1}, \qquad \nabla_\theta J = \sum_{s=0}^{S-1} B_s^\top \lambda_{s+1}.$$

$\lambda_s \in \mathbb{R}^n$ is the cost sensitivity to x_s. Explicit control costs or control-dependent initialization add their own terms. Neither is introduced here.

Minimum sending/receiving laws, queue caps, active constraints, shocks and leader interactions can be nonsmooth. Smooth-branch sensitivities do not settle event boundaries. A hybrid reference may require event-time terms; holding an event fixed can yield a different derivative. No unapproved smoothing or private stochastic resampling resolves this issue. An existence theorem is not a differentiability or useful-gradient theorem. Mean-capacity parameter sensitivities are different from the ramp-rate TST derivative.

Three evidence levels remain distinct: a justified derivative for a bound task; numerical/reference agreement; and useful common perturbations in a richer independent reference. This review gives the conditional general path, not the latter two. The unresolved history/action contract already prevents positive core-task admission.

Provenance. Source versions, selected sections, URLs and access limits are in `extraction/rm10/primary_source_audit.md`. The task/consistency definitions, conditional sufficiency statements and TST chain-rule derivations are reviewer constructions. They impose no new traffic equation on the repository.